Xiongwei Luo

Zurich, Switzerland | +86 137 8924 2010 | xiongweiluo1@gmail.com | GitHub
AI Product and Application Development | Internship / Working Student

EDUCATION

ETH Zurich Zurich, Switzerland
MSc in Integrated Building Systems *Sep 2025 - Jul 2027 (expected)*
Relevant coursework: Machine Learning, Hands-On Deep Learning, Scientific Machine Learning, Building Agentic AI Systems.

Tongji University Shanghai, China
Bachelor of Architecture *Sep 2019 - Jul 2025*
Studies in urban data analysis and human-computer interaction; academic scholarships in four consecutive years.

National University of Singapore Singapore
Young Fellowship Program | Summer programme on interdisciplinary AI *Jul 2025*

TECHNICAL SKILLS

Programming:	Python, TypeScript/JavaScript, SQL, GDScript
Frameworks and tools:	React, FastAPI, Node.js, Docker, Git, GitHub Actions, Figma
AI and agents:	MCP/FastMCP, Function Calling, JSON Schema, QLoRA, paired evaluation
Data and systems:	MySQL, SQLite, Redis, Celery, WebSocket, automated testing

PROFESSIONAL EXPERIENCE

Product Design and Prototyping Intern
JD.com | Autonomous Vehicle Product Team *Jun 2026 - Sep 2026*
- Built a unified B2B prototype with **17 top-level menus and 91 navigation entries**; integrated 54 entries with existing business prototypes and documented missing requirements for the remaining 37.
- Implemented deep links, history navigation and unsaved-form protection. Established **91-page screenshot audits** and interaction regression checks; delivered a self-contained HTML export for offline demonstrations.

PROJECTS

RhinoCoder | Verifiable AI Agent for 3D Modelling | GitHub
Independent project | Python, React, TypeScript, FastMCP, RhinoCommon, SQLite
- Built a planning, execution and verification loop using **23 versioned MCP tools**, Rhino scene inspection and geometric assertions. Added idempotency keys, cancellation, undo, task rollback and privacy checks.
- Passed **270/270 valid real-Rhino runs** on a saturated fixed benchmark: 30 tasks, three configurations and three repetitions. Reported the limits of generalising beyond this benchmark.
- Used **QLoRA** to improve exact tool-call sequence matching from **20/80 to 77/80** on 80 offline task families. The real-Rhino quality gate remains unmet; the default production route is unchanged.

LabelHub | AI Annotation and Quality Platform | Mock demo
Team project | Schema architecture and core development | React, FastAPI, MySQL, Redis, Celery
- Built shared schema contracts and a designer-compiler-renderer pipeline. Applied deterministic preflight checks to field suggestions; used **Function Calling for asynchronous AI pre-review** and reviewer diffs for oversight.
- Implemented asynchronous pre-review and exports with idempotency, retries and status tracking. Delivered Docker Compose deployment and layered CI, with **nine real-backend E2E scenarios passing**.

Family Garden | Collaborative AI Game | Live demo
Team project | AI, backend and game systems | Godot, GDScript, Node.js, CloudBase, WebSocket
- Built AI memory and image-analysis workflows with schema validation and content moderation. Implemented member identity, family data isolation, private photo storage and WebSocket presence.
- Used server-side transactions and operation identifiers for inventory changes, planting and harvesting. Shipped a Web release validated by **26/26 regression checks**, **4/4 production E2E scenarios** and iPhone 16 Safari checks.

ADDITIONAL

Languages: Chinese (native), English (IELTS 7.0), German (B2)